Formula sheet for Quiz 10

Position

Token at position k: $x_k = E_{w_k} + P_k$.
Sinusoidal, with $n = 10{,}000$:

$$P(k, 2i) = \sin\Big(\frac{k}{n^{2i/d}}\Big), \qquad P(k, 2i+1) = \cos\Big(\frac{k}{n^{2i/d}}\Big)$$

Learned: P is an $L \times d$ table, one row per position.

The block

Post-norm, the PyTorch default:

$$\mathbf{x} \leftarrow \mathrm{LayerNorm}(\mathbf{x} + \mathrm{Attn}(\mathbf{x})), \qquad \mathbf{x} \leftarrow \mathrm{LayerNorm}(\mathbf{x} + \mathrm{FFN}(\mathbf{x}))$$

Pre-norm:

$$\mathbf{x} \leftarrow \mathbf{x} + \mathrm{Attn}(\mathrm{LayerNorm}(\mathbf{x})), \qquad \mathbf{x} \leftarrow \mathbf{x} + \mathrm{FFN}(\mathrm{LayerNorm}(\mathbf{x}))$$

$\mathrm{FFN}(\mathbf{x}) = W_2\,\varphi(W_1\mathbf{x} + b_1) + b_2$, with W_1 of shape $d \times 4d$ and W_2 of shape $4d \times d$.

Normalization

One token $\mathbf{x} \in \mathbb{R}^d$. LayerNorm:

$$\mu = \frac{1}{d}\sum_i x_i, \quad \sigma^2 = \frac{1}{d}\sum_i (x_i - \mu)^2, \quad \mathbf{y} = \gamma \odot \frac{\mathbf{x} - \mu}{\sqrt{\sigma^2 + \varepsilon}} + \beta$$

BatchNorm uses the same affine form, with μ and σ^2 taken across the batch for each coordinate.

Attention

$$\mathbf{A}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = S\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + M\right)\mathbf{V}$$

S is softmax, so each row of the weight matrix sums to 1. A masked entry of M is $-\infty$. Causal: $M_{t,t'} = -\infty$ when $t' > t$.

Sampling and memory

$$p_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

$T = 1$ is plain softmax. Top-k keeps the k largest p_i and renormalizes. Top-p keeps the smallest set whose probabilities sum to at least p, and renormalizes.
Weight memory = (number of parameters) × (bytes per parameter). fp32: 4, fp16 and bf16: 2, int8: 1, int4: 1/2.

Embeddings and LoRA

Mean pool, then normalize: $\mathbf{v} = \dfrac{\sum_t m_t \mathbf{h}_t}{\sum_t m_t}$, then $\mathbf{v} \leftarrow \mathbf{v}/\|\mathbf{v}\|$. Cosine of two normalized vectors is their dot product.

$$y = Wx + BAx, \qquad A \in \mathbb{R}^{r \times d},\ B \in \mathbb{R}^{d \times r}$$

Frozen count d^2. LoRA trained count $2dr$, fraction $2r/d$. Training W itself instead: trained count d^2.